

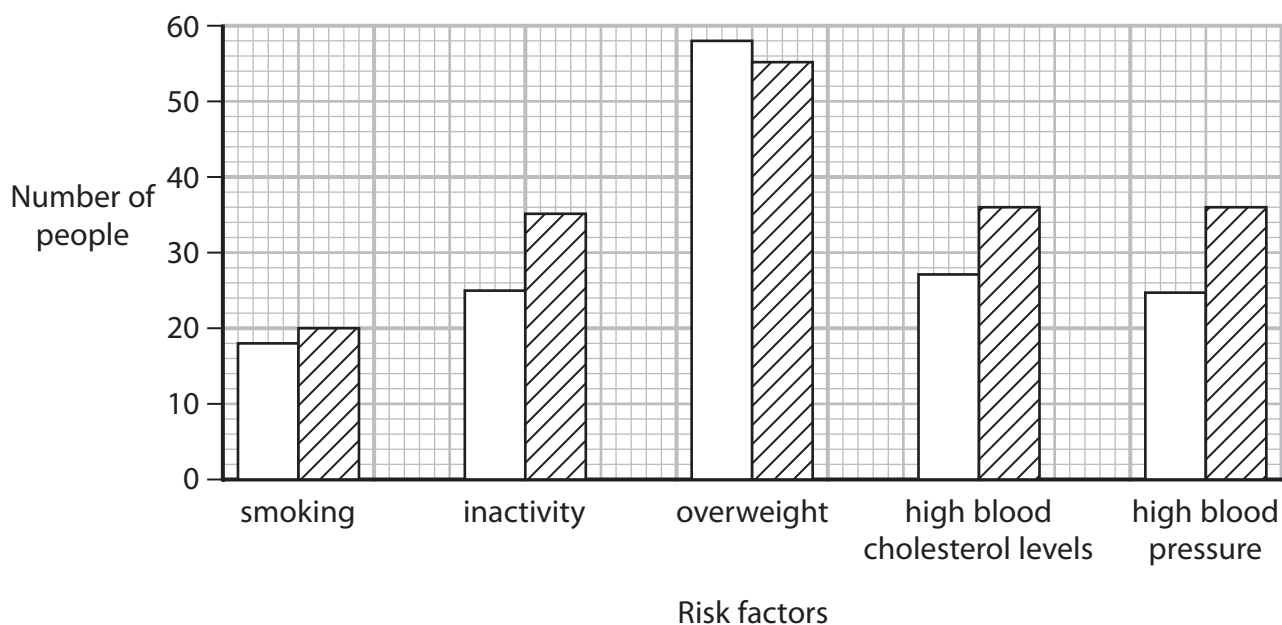
- 6 The risk of developing cardiovascular disease (CVD) is dependent on a number of factors.

- (a) People's perception of these risks are often very different from the actual risks.

A study was designed to compare perceived risks with actual risks.

A group of people were asked if they thought they were at risk (perceived risk) of CVD. The same group of people were then examined to determine their actual risks of CVD.

The graph shows the results of this study.



Key:



number of people with a perceived risk



number of people with an actual risk

- (i) Describe **two** conclusions that can be drawn from this study.

(2)

1

.....

.....

2

.....

.....



(ii) Explain how this study should have been designed to allow valid conclusions to be made.

(3)

(iii) Explain why it is important that a person's perception of a risk is close to the actual risk.

(3)

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- (b) The 10-year risk of a person developing CVD can be estimated in a number of ways.

The Framingham Risk Score uses a point scoring system to estimate the 10-year risk of developing CVD.

Table 1 shows the point scores for different risk factors in women.

Table 1

Risk factor	Points per age group			
	35 to 39	40 to 44	45 to 49	50 to 54
age	−3	0	+3	+6
total cholesterol level / mg dm^{−3}				
<160	0	0	0	0
160 to 199	+4	+3	+3	+2
200 to 239	+8	+6	+6	+4
240 to 279	+11	+8	+8	+5
smoker	+9	+7	+7	+4
systolic blood pressure / mm Hg				
<120	0	0	0	0
120 to 129	+1	+1	+1	+1
130 to 139	+2	+2	+2	+2

Table 2 shows the 10-year risk for the total points scored, calculated using the information in Table 1.

Table 2

Total points scored	10-year risk of developing CVD (%)
16	4
17	5
18	6
19	8
20	11
21	14
22	17



(i) Calculate the 10-year risk of CVD for a 35-year-old woman who:

- has a total cholesterol level of 242 mg dm^{-3}
- is a smoker
- has a systolic blood pressure of 132 mm Hg.

(2)

Answer %

(ii) Explain the lifestyle changes this woman could make to reduce her 10-year risk of CVD.

Use the information in the table to support your answer.

(2)

(iii) Which pair of drugs may reduce the 10-year risk of CVD for this woman?

(1)

- ☐ A anticoagulants and statins
- ☐ B anticoagulants and antihypertensives
- ☐ C antihypertensives and platelet inhibitors
- ☐ D antihypertensives and statins

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(iv) Suggest why this method of determining the 10-year risk of CVD may not be accurate.

(2)

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(Total for Question 6 = 15 marks)

